

자연어처리 기말 프로젝트

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GPT-2 classifier.py 감성 분석

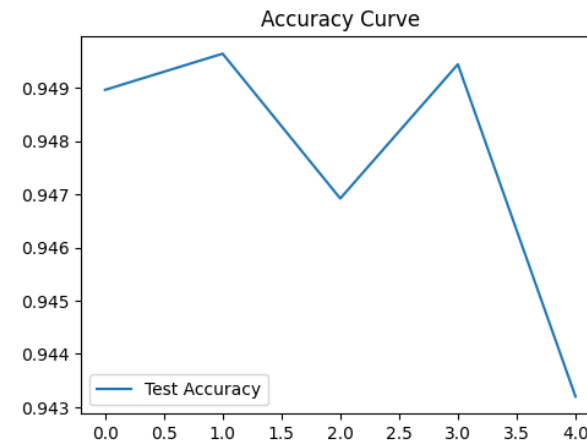
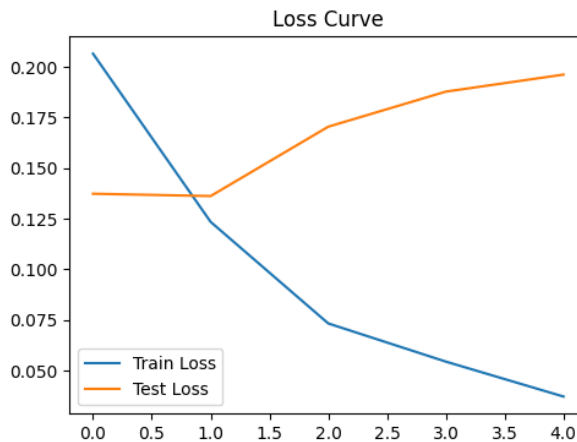
- 데이터셋: SST, CFIMDB
- 결과 (dev set):
 - 1) SST dev set
 - Accuracy: 44.8%
 - 2) CFIMDB dev set
 - Accuracy: 83.7%

```
DONE Test  
dev acc :: 0.448
```

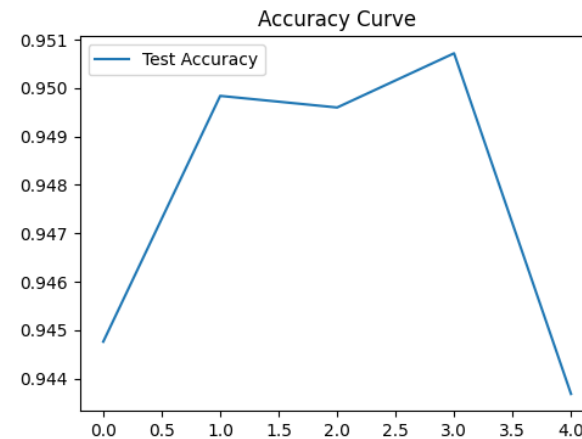
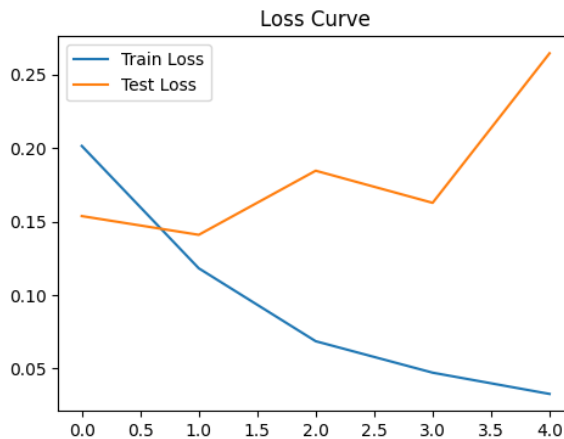
```
DONE Test  
dev acc :: 0.837
```

가중 앙상블 모델 감성 분석

- 데이터셋: Stanford IMDB
- RoBERTa 단일 모델
 - Test Accuracy : 94.96%
 - Test Loss : 0.1362



- XLNet 단일 모델
 - Test Accuracy: 95.07%
 - Test Loss: 0.1410



가중 앙상블 모델 감성 분석

- RoBERTa + XLNet 앙상블 모델

```
Best Weights: {'roberta': 0.5159372768808733, 'xlnet': 0.4840627231191267}
```

```
Final Train (5k) -> Loss 0.3323 | Acc 0.9952
```

```
Final Test      -> Loss 0.3773 | Acc 0.9401
```

```
Classification Report:
```

	precision	recall	f1-score	support
negative	0.93	0.95	0.94	12500
positive	0.95	0.93	0.94	12500
accuracy			0.94	25000
macro avg	0.94	0.94	0.94	25000
weighted avg	0.94	0.94	0.94	25000

소네트 생성(Sonnet Generation)

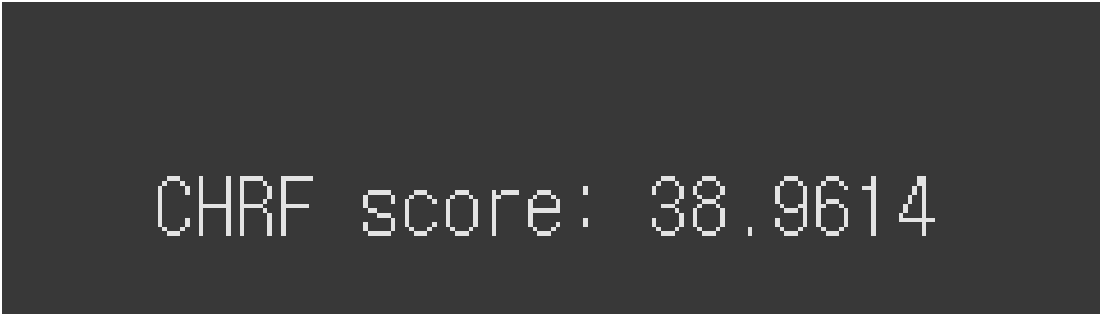
- 데이터셋: sonnets.txt, sonnets_held_out.txt

- 전체 파라미터 미세조정: 26점



```
CHRF score: 26.2012
```

- LoRA, PEFT 적용: 39점



```
CHRF score: 38.9614
```

의역 탐지(Paraphrase Detection)

- 데이터셋: Quora dev set
- Accuracy: 89.7%
- F1 Score: 0.8896

```
Loaded 40429 train examples from data/quora-dev.csv  
eval: 100%|██████████| 5054/5054 [01:20<00:00, 63.00it/s]  
Dev accuracy: 0.8966, F1 score: 0.8896
```